

Mechanical Waves and Sound

How disturbances travel: wave speed, energy, interference, standing waves, the Doppler effect, and how ultrasound sees inside the body.

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A wave carries energy and information from one place to another **without carrying matter** along with it. Stadium crowds, ripples on a pond, a plucked guitar string, sound, earthquakes: in each, the medium oscillates locally while a pattern travels. This chapter builds the mathematics of travelling waves, finds what sets their speed, and explores what happens when waves meet: interference, standing waves, and beats. It ends with the Doppler effect and medical ultrasound, among the most widely used applications of wave physics in medicine.

YOU WILL BE ABLE TO

- Distinguish transverse from longitudinal waves, and wave speed from particle velocity.
- Use $y = A \sin(kx - \omega t)$ with $v = f\lambda = \omega/k$.
- Derive and use the wave speeds on a string, $\sqrt{F/\mu}$, and of sound in fluids and solids.
- Calculate wave intensity, the inverse-square law and sound levels in decibels.
- Apply superposition to interference, reflection, standing waves on strings and in pipes, and beats.
- Use the Doppler effect for sound, including Doppler ultrasound, and understand shock waves.
- Explain ultrasound imaging using acoustic impedance and reflection.

THINK FIRST

A doctor's Doppler ultrasound machine sends 5 MHz sound into the body, far above human hearing, yet the machine makes an audible “whooshing” sound that pulses with the heartbeat. Where does the audible sound come from?

From the **Doppler shift**. Sound reflected from moving red blood cells comes back at a slightly different frequency. For blood moving at a few tens of cm/s, the shift is about 1–3 kHz (Example 11.7), right in the middle of human hearing. The machine electronically extracts the *difference* between the transmitted and received frequencies and plays it through a speaker. Faster flow gives a higher pitch, so the sound rises and falls with each heartbeat.

§11.1

What is a wave?

A **mechanical wave** is a disturbance that travels through a medium (string, air, water, rock), with each part of the medium oscillating about its equilibrium position. Electromagnetic waves (Chapter 21) need no medium.

- In a **transverse** wave, the particles move *perpendicular* to the direction of travel. Examples: waves on a string, and S-waves in earthquakes.
- In a **longitudinal** wave, the particles move *parallel* to the direction of travel, creating compressions and rarefactions. Examples: sound in air, and P-waves in earthquakes.

Water surface waves are a mixture: the water particles move in circles.

§11.2

Describing a travelling wave

A pulse with shape $y = f(x)$ at $t = 0$ that moves in the $+x$ direction at speed v keeps its shape. At time t it is $y = f(x - vt)$. A wave travelling in the $-x$ direction is $f(x + vt)$.

For a **sinusoidal** wave travelling in the $+x$ direction:

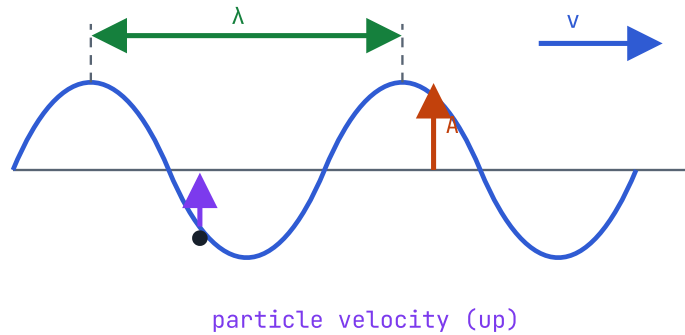
$$y(x, t) = A \sin(kx - \omega t),$$

- A is the amplitude.

- $k = 2\pi/\lambda$ is the **wave number** (rad/m), where λ is the **wavelength**.
- $\omega = 2\pi f$ is the angular frequency, where f is the frequency and $T = 1/f$ the period.

A crest moves a distance λ in one period T , so the wave speed is

$$v = \frac{\lambda}{T} = f\lambda = \frac{\omega}{k}.$$



A sinusoidal wave at one instant. The pattern moves right at the wave speed v . Each bit of the string only moves up and down.

COMMON MISTAKE

The **wave speed** v is the speed of the *pattern*. It is set by the properties of the medium. The **particle velocity** $\partial y / \partial t = -A\omega \cos(kx - \omega t)$ is how fast each bit of the medium moves. Its maximum, $A\omega$, depends on the amplitude and frequency. The two are completely different quantities.

Every travelling wave satisfies the **wave equation**:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}.$$

Deriving this equation from Newton's laws for a particular medium tells us the wave speed.

WORKED EXAMPLE 11.1

Reading a wave function

A wave on a string is described by $y = 0.020 \sin(4\pi x - 100\pi t)$, with y and x in metres and t in seconds. Find the amplitude, wavelength, frequency, wave speed and direction, and the maximum speed of a point on the string.

Solution

- $A = 0.020$ m.
- $k = 4\pi$, so $\lambda = 2\pi/k = 0.50$ m.
- $\omega = 100\pi$, so $f = 50$ Hz.
- $v = \omega/k = 100\pi/4\pi = 25$ m/s, in the $+x$ direction, because of the form $kx - \omega t$.
- Maximum particle speed: $A\omega = 0.020(100\pi) = 6.3$ m/s.

§11.3

Wave speeds

Transverse waves on a string with tension F and mass per unit length μ :

$$v = \sqrt{\frac{F}{\mu}}.$$

This was predicted by dimensional analysis in Problem P1.6, and is derived in Problem P11.13.

Sound (longitudinal waves) travels at a speed set by a stiffness divided by an inertia:

$$v = \sqrt{\frac{\text{elastic modulus}}{\text{density}}} : \quad v_{\text{fluid}} = \sqrt{\frac{B}{\rho}}, \quad v_{\text{rod}} = \sqrt{\frac{Y}{\rho}}.$$

For an **ideal gas**, the compressions are fast and adiabatic, which gives

$$v = \sqrt{\frac{\gamma RT}{M}},$$

where $\gamma = 1.40$ for air (Chapter 12), $R = 8.314$ J/(mol K), T is the absolute temperature and M the molar mass. The speed of sound grows as $\sqrt{T}$ and does not depend on pressure.

MEDIUM	SPEED OF SOUND (M/S)
Air, 0 °C	331
Air, 20 °C	343
Helium, 20 °C	1007

MEDIUM	SPEED OF SOUND (M/S)
Water	1480
Soft tissue (average)	1540
Bone	3000–4000
Steel	5900 (longitudinal, bulk)

WORKED EXAMPLE 11.2**Tuning a guitar string**

A guitar's A-string is 0.65 m long between its fixed ends and has a mass of 3.0 g per metre. What tension makes its fundamental frequency 110 Hz?

Solution

In its fundamental mode, the string is half a wavelength long (see the section on standing waves below), so $\lambda = 2L = 1.30$ m.

$$v = f\lambda = 110(1.30) = 143 \text{ m/s}, \quad F = \mu v^2 = (3.0 \times 10^{-3})(143)^2 = 61 \text{ N}.$$

Evaluate. The six strings together pull on the guitar's neck with roughly 400–700 N, which is why necks contain a steel truss rod. Tuning a string by turning its peg changes F , and $f \propto \sqrt{F}$.

§11.4

Energy, intensity and decibels

A wave carries energy. For a sinusoidal wave on a string, the average power transmitted is

$$\bar{P} = \frac{1}{2} \mu v \omega^2 A^2.$$

It is proportional to the **square of the amplitude** and the **square of the frequency**. The same scalings hold for all mechanical waves.

The **intensity** I of a wave in three dimensions is the power per unit area perpendicular to its direction of travel, in W/m². A point source radiating power P equally in all directions spreads

that power over spheres of area $4\pi r^2$:

$$I = \frac{P}{4\pi r^2} \quad (\text{inverse-square law}).$$

The ear responds to an enormous range of intensities, from about 10^{-12} W/m² (just audible at 1 kHz) to about 1 W/m² (painful). So we use a logarithmic scale, the **sound intensity level**:

$$\beta = (10 \text{ dB}) \log_{10} \frac{I}{I_0}, \quad I_0 = 10^{-12} \text{ W/m}^2.$$

- Every factor of 10 in intensity adds 10 dB.
- Doubling the intensity adds 3 dB.
- Doubling the distance from a point source subtracts 6 dB.

SOUND	LEVEL (DB)
Threshold of hearing	0
Quiet library	30–40
Conversation	60
Heavy traffic	80–85
Rock concert, chainsaw	110
Threshold of pain	120–130

WORKED EXAMPLE 11.3

How loud is the loudspeaker?

A loudspeaker radiates 10 W of sound uniformly in all directions. Find the intensity and sound level at (a) 5.0 m and (b) 10 m. (c) What is the level at 5.0 m if a second identical speaker is added beside the first?

Solution

(a) At 5.0 m:

$$I = \frac{10}{4\pi(5.0)^2} = 0.032 \text{ W/m}^2, \quad \beta = 10 \log_{10} \frac{0.032}{10^{-12}} = 10(10.5) = 105 \text{ dB}.$$

(b) Doubling the distance quarters the intensity: $\beta = 105 - 6 = 99$ dB.

(c) Doubling the power doubles the intensity: $105 + 3 = 108$ dB, not 210 dB.

IN PRACTICE

Noise and hearing loss. Workplace safety rules typically limit exposure to 85 dB averaged over 8 hours. Every 3 dB increase halves the allowed exposure time, so 88 dB is allowed for 4 hours, 91 dB for 2 hours, and 100 dB for only about 15 minutes. Prolonged loud sound damages the hair cells of the cochlea, which do not regrow, causing permanent noise-induced hearing loss that typically appears first near 4 kHz. Audiologists measure hearing thresholds in dB at each frequency to produce an *audiogram*.

§11.5

Superposition and interference

KEY IDEA

Principle of superposition. When two or more waves overlap in a linear medium, the resulting displacement is the sum of their individual displacements. The waves pass through each other unchanged.

Two waves of equal frequency arriving at a point:

- **in phase** (crest meets crest) add: **constructive interference**;
- **half a cycle out of phase** (crest meets trough) cancel: **destructive interference**.

If two sources oscillate in phase, the result at a given point depends on the **path difference** Δr from the two sources:

$$\Delta r = n\lambda \Rightarrow \text{constructive}, \quad \Delta r = \left(n + \frac{1}{2}\right)\lambda \Rightarrow \text{destructive}.$$

Noise-cancelling headphones use destructive interference: they generate an inverted copy of the incoming noise.

Reflection. A pulse reflected from a **fixed end** comes back inverted, with a phase change of π . A pulse reflected from a **free end** comes back upright. At a junction between two media, part of the

wave is reflected and part transmitted. The fraction reflected depends on how different the media are, which is the key to ultrasound imaging below.

§11.6

Standing waves

Two identical waves travelling in opposite directions superpose to form a **standing wave**:

$$y = A \sin(kx - \omega t) + A \sin(kx + \omega t) = 2A \sin kx \cos \omega t.$$

Every point oscillates in step, with an amplitude $2A|\sin kx|$ that depends on its position.

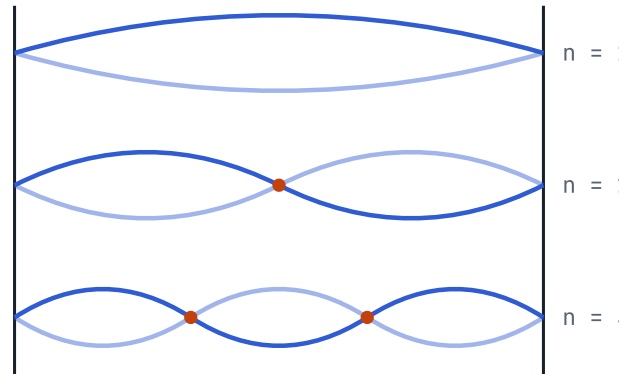
- **Nodes**, where the amplitude is zero, occur at $x = 0, \lambda/2, \lambda, \dots$
- **Antinodes**, where the amplitude is $2A$, lie halfway between the nodes.
- The distance between neighbouring nodes is $\lambda/2$.
- Energy does not travel along a standing wave.

Strings fixed at both ends

Both ends must be nodes, so a whole number of half-wavelengths must fit in the length L :

$$L = n \frac{\lambda_n}{2} \Rightarrow f_n = \frac{nv}{2L} = \frac{n}{2L} \sqrt{\frac{F}{\mu}}, \quad n = 1, 2, 3, \dots$$

The **fundamental**, or first harmonic, is $n = 1$. The higher **harmonics** are whole-number multiples of it. A plucked string vibrates in a mixture of harmonics, and the particular mixture gives an instrument its characteristic tone, or *timbre*.



Standing waves on a string fixed at both ends. The orange dots mark interior nodes. The n th harmonic has n loops and frequency n times the fundamental.

Air columns

Sound standing waves form in pipes. At a **closed** end the air cannot move, so there is a **displacement node**. At an **open** end there is a displacement antinode, which is also a pressure node.

- **Open at both ends:** $f_n = \frac{nv}{2L}$, for $n = 1, 2, 3, \dots$ (all harmonics). Examples: the flute, and open organ pipes.
- **Closed at one end:** $L = \frac{n\lambda}{4}$ with n odd only, so $f_n = \frac{nv}{4L}$ for $n = 1, 3, 5, \dots$. The fundamental is an octave lower than for an open pipe of the same length. Examples: the clarinet, and stopped organ pipes.

WORKED EXAMPLE 11.4

Why we hear best near 3 kHz

The ear canal is a tube about 2.5 cm long, open at the outer end and closed by the eardrum. Find its fundamental resonant frequency.

Solution

Treat it as a pipe closed at one end:

$$f_1 = \frac{v}{4L} = \frac{343}{4(0.025)} = 3.4 \times 10^3 \text{ Hz.}$$

Evaluate. The resonance boosts sound pressure at the eardrum by roughly 10–15 dB around 3–4 kHz. This is exactly where human hearing is most sensitive, and it is also

where noise damage tends to appear first. This frequency range carries many of the consonant sounds of speech, which is why high-frequency hearing loss makes speech hard to follow.

§11.7

Beats

Two waves of slightly different frequencies, f_1 and f_2 , superpose to give a tone at the average frequency whose loudness rises and falls. Adding the two sinusoids:

$$\cos 2\pi f_1 t + \cos 2\pi f_2 t = 2 \cos \left(2\pi \frac{f_1 - f_2}{2} t \right) \cos \left(2\pi \frac{f_1 + f_2}{2} t \right).$$

The slowly varying factor is the envelope, and the loudness peaks twice per cycle of that envelope. So the **beat frequency** is

$$f_{\text{beat}} = |f_1 - f_2|.$$

WORKED EXAMPLE 11.5

Tuning by beats

A piano tuner strikes a 440 Hz tuning fork together with a piano string and hears 3 beats per second. When she tightens the string slightly, the beats slow down. What was the string's original frequency?

Solution

The beat frequency of 3 Hz means the string was at either 437 Hz or 443 Hz. Tightening a string raises its frequency, since $f \propto \sqrt{F}$. Raising 443 Hz would *increase* the beat rate, while raising 437 Hz towards 440 Hz *decreases* it. So the string was at **437 Hz**.

§11.8

The Doppler effect

When a source of sound and an observer move relative to the air, the observed frequency differs from the emitted frequency. For a source moving at speed v_s and an observer moving at speed v_o

along the line joining them, with the speed of sound v :

$$f' = f \frac{v \pm v_o}{v \mp v_s}.$$

Use the **upper signs** when the motion is **towards** the other party. Approaching raises the frequency, and receding lowers it.

- A **moving source** compresses the wavefronts ahead of it and stretches them behind it, so the wavelength changes.
- A **moving observer** meets the wavefronts more or less often, but the wavelength is unchanged.

These two effects are not quite symmetric, which is why v_s appears in the denominator and v_o in the numerator.

WORKED EXAMPLE 11.6

A passing ambulance

An ambulance siren emits 700 Hz. The ambulance travels at 25 m/s past a pedestrian standing still. What frequencies does the pedestrian hear as it approaches and as it recedes? Take $v = 343$ m/s.

Solution

$$f_{\text{approach}} = 700 \cdot \frac{343}{343 - 25} = 755 \text{ Hz}, \quad f_{\text{recede}} = 700 \cdot \frac{343}{343 + 25} = 652 \text{ Hz}.$$

The pitch drops by about 100 Hz, a 15% fall, as the ambulance passes.

WORKED EXAMPLE 11.7

Measuring blood flow with Doppler ultrasound

A 5.0 MHz ultrasound beam is aimed at an artery, making an angle of 60° with the direction of blood flow. The echo from the moving red blood cells is shifted by 1.3 kHz. Find the speed of the blood. The speed of sound in tissue is 1540 m/s.

Solution

Double shift. The cells first act as *moving observers* of the incoming beam. They then re-emit the sound as *moving sources*. Only the component of the blood velocity along the beam, $v \cos \theta$, matters. For $v \ll c$, the two shifts add to give

$$\Delta f \approx \frac{2f_0 v \cos \theta}{c}.$$

Solving for the blood speed:

$$v = \frac{c \Delta f}{2f_0 \cos \theta} = \frac{1540(1300)}{2(5.0 \times 10^6)(0.50)} = 0.40 \text{ m/s.}$$

Evaluate. The shift is about 1 kHz, right in the audible range, as the “Think first” box described. Clinicians keep the angle below about 60° . Near 90° , $\cos \theta \rightarrow 0$, so the shift vanishes, and a small error in the angle produces a large error in the speed (Problem P11.15).

Shock waves. If a source moves faster than the waves it creates ($v_s > v$), the wavefronts pile up into a cone, a **shock wave**, with half-angle given by $\sin \theta = v/v_s = 1/M$, where M is the **Mach number**. A supersonic aircraft drags this cone along behind it, and the sonic boom is heard as the cone sweeps past the ground. Boat wakes are the same phenomenon in water.

§11.9

Ultrasound imaging

Medical ultrasound uses frequencies of 2–15 MHz. The wavelength in tissue, $\lambda = c/f$, sets the finest detail that can be seen: at 5 MHz, $\lambda = 1540/5 \times 10^6 = 0.31 \text{ mm}$. Higher frequencies give sharper images, but they are absorbed more strongly, so they cannot see as deep.

Images are built from **echoes**. A pulse is sent in, and the time t for its echo to return gives the depth of the reflecting boundary, $d = ct/2$. The strength of the echo depends on the **acoustic impedance** $Z = \rho c$ of the tissues on either side of the boundary. For a wave arriving perpendicular to the boundary, the fraction of the intensity reflected is

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2.$$

MATERIAL	Z ($\times 10^6 \text{ KG M}^{-2} \text{ S}^{-1}$, “MRAYL”)
Air	0.0004
Fat	1.38
Soft tissue / water	1.5–1.7
Bone	6–8

WORKED EXAMPLE 11.8**Why ultrasound needs gel**

Find the fraction of ultrasound intensity reflected at (a) a soft tissue–air boundary and (b) a soft tissue–bone boundary. Take $Z_{\text{tissue}} = 1.63$, $Z_{\text{air}} = 0.0004$ and $Z_{\text{bone}} = 7.8$ (in MRayl).

Solution

(a) Tissue–air:

$$R = \left(\frac{1.63 - 0.0004}{1.63 + 0.0004} \right)^2 = 0.9990.$$

So 99.9% of the sound is reflected.

(b) Tissue–bone:

$$R = \left(\frac{7.8 - 1.63}{7.8 + 1.63} \right)^2 = (0.654)^2 = 0.43.$$

Evaluate. Any air gap between the probe and the skin would reflect almost all the sound before it entered the body, so a water-based gel is used to couple the probe to the skin. Air and bone also cast acoustic “shadows”. That is why ultrasound cannot image through the lungs or the adult skull, and why it is ideal for soft tissues: fetuses, the heart, the liver and blood vessels. It uses no ionising radiation and works in real time, which is why it is often the first imaging test.

IN PRACTICE

Waves across science and engineering.

- **Seismology.** Earthquakes send out P-waves (longitudinal, faster) and S-waves (transverse, slower). The delay between their arrivals gives the distance to the source (Problem P11.18). Because S-waves cannot travel through liquids, which have no shear stiffness, their absence on the far side of the Earth revealed that the outer core is liquid.
- **Non-destructive testing.** Engineers send ultrasonic pulses through welds, aircraft wings and rails. Echoes from internal cracks reveal flaws before they cause failure.
- **Lithotripsy.** Focused shock waves, generated outside the body, break up kidney stones into fragments small enough to pass naturally, with no surgical incision.
- **Sonar** and the echolocation of bats and dolphins use the same echo timing and Doppler shifts as medical ultrasound.

CHAPTER SUMMARY

- Travelling wave $y = A \sin(kx - \omega t)$, with $k = 2\pi/\lambda$, $\omega = 2\pi f$ and $v = f\lambda = \omega/k$. The particle speed is $\partial y / \partial t$, which is not v .
- String: $v = \sqrt{F/\mu}$. Sound: $v = \sqrt{B/\rho}$ in fluids, $\sqrt{Y/\rho}$ in rods, and $\sqrt{\gamma RT/M}$ in gases (343 m/s in air at 20 °C).
- Power $\propto A^2 \omega^2$. Point source: $I = P/4\pi r^2$. Sound level $\beta = 10 \log(I/I_0)$ dB.
- Superposition: constructive interference when $\Delta r = n\lambda$, destructive when $\Delta r = (n + \frac{1}{2})\lambda$. Reflection from a fixed end inverts the wave.
- Standing waves: string or open pipe, $f_n = nv/2L$; pipe closed at one end, $f_n = nv/4L$ with n odd.
- Beats: $f_{\text{beat}} = |f_1 - f_2|$.
- Doppler: $f' = f(v \pm v_o)/(v \mp v_s)$, with upper signs for approach. Doppler ultrasound: $\Delta f \approx 2f_0 v \cos \theta / c$. Mach cone: $\sin \theta = 1/M$.
- Ultrasound: depth $d = ct/2$, impedance $Z = \rho c$, and reflected fraction $R = ((Z_2 - Z_1)/(Z_2 + Z_1))^2$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P11.1

Explain the difference between the wave speed and the particle speed for a wave on a string. Can the maximum particle speed ever exceed the wave speed? What would that require?

P11.2

Steel is about 6500 times denser than air, yet sound travels about 17 times faster in steel. How can this be?

P11.3

A sound of 90 dB is described as “a bit louder” than one of 80 dB. By what factor does its intensity differ? Why does this matter for hearing safety?

P11.4

Breathing helium makes a person’s voice sound high-pitched and squeaky, yet the vocal cords vibrate at almost the same frequency. Explain what changes.

LEVEL B – STANDARD PROBLEMS

P11.5

A wave is described by $y = 0.050 \sin(5\pi x + 50\pi t)$, with SI units. Find its wavelength, frequency, speed and direction of travel, and the maximum transverse speed of a point on the string.

P11.6

A steel guitar string ($\rho = 7800 \text{ kg/m}^3$) of diameter 0.25 mm and vibrating length 0.648 m is tuned to 330 Hz (E4). Find its mass per unit length and the required tension.

P11.7

You see a lightning flash and hear the thunder 3.0 s later. How far away was the strike? Explain the common rule of thumb “count the seconds and divide by three to get kilometres”.

P11.8

A jackhammer produces a sound level of 100 dB at 2.0 m. (a) What is the level at 20 m? (b) What acoustic power does it radiate, assuming it radiates equally in all directions? (c) How close can a worker stand while staying at or below 85 dB?

P11.9

What length of organ pipe gives a fundamental of 65.4 Hz (the note C₂) if the pipe is (a) open at both ends and (b) closed at one end? What is the second resonant frequency of each?

P11.10

Two violin strings are tuned to 440 Hz and 444 Hz. By what percentage should the tension in the higher string be changed to eliminate the beats?

P11.11

A bat flying at 10 m/s towards a wall emits a 40.0 kHz call. What frequency does the bat hear in the echo from the wall? (Take $v = 343$ m/s.)

P11.12

A 3.5 MHz ultrasound probe is used in an abdominal scan. (a) Find the wavelength in soft tissue. (b) An echo returns 65 μ s after the pulse is sent. How deep is the reflecting boundary? (c) What fraction of the intensity is reflected at a fat-muscle boundary, with $Z_{\text{fat}} = 1.38$ MRayl and $Z_{\text{muscle}} = 1.70$ MRayl? Why are such weak echoes still useful?

LEVEL C – CHALLENGE PROBLEMS

P11.13

Speed of a wave on a string. A pulse travels at speed v along a string with tension F and mass per unit length μ . Work in the frame moving with the pulse, in which the string flows through a stationary pulse shape at speed v . Consider a short segment at the top of the pulse, approximated as an arc of a circle of radius R subtending angle 2θ . By applying Newton's second law to this segment, show that $v = \sqrt{F/\mu}$.

P11.14

A uniform rope of length L hangs freely from a ceiling. (a) Show that the speed of a transverse pulse at a height y above the lower end is $v = \sqrt{gy}$. (b) Show that the time for a pulse to travel the full length of the rope is $2\sqrt{L/g}$. (c) Evaluate this for a 5.0 m rope.

P11.15

Doppler error analysis. A 4.0 MHz Doppler probe makes an angle of 45° with an artery and records a shift of 2.0 kHz. (a) Find the blood speed. (b) Show that a small error $\delta\theta$ in the assumed angle produces a fractional error in speed of $\delta v/v = \tan\theta \delta\theta$. (c) Find the percentage error caused by a 5° error in angle at 30° , 60° and 80° , and explain the clinical guideline to keep the angle at or below 60° .

P11.16

(a) Show that the superposition of $A \sin(kx - \omega t)$ and $A \sin(kx + \omega t)$ is $2A \sin kx \cos \omega t$, and locate its nodes and antinodes. (b) A string 1.20 m long, fixed at both ends, vibrates in its third harmonic at 150 Hz. Find the wave speed, the positions of all the nodes, and the frequency of the fundamental.

P11.17

Sonic boom. A jet flies horizontally at Mach 2.0 at an altitude of 10 km. Take the speed of sound as 340 m/s. (a) Find the half-angle of the Mach cone. (b) How long after the jet passes directly overhead does an observer on the ground hear the boom? (c) How far has the jet travelled in that time?

P11.18

Locating an earthquake. P-waves travel through the crust at about 6.0 km/s and S-waves at about 3.5 km/s. A seismometer records the S-waves arriving 20 s after the P-waves. (a) How far away is the earthquake? (b) How many seismometers are needed to pinpoint its location, and why? (c) Explain how the absence of S-waves on the side of the Earth opposite an earthquake shows that the Earth's outer core is liquid.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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